Robotics, Lab assignment 2

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Due on: 13-11-2026, 23:59:59
To be addressed in groups of up to 3 students.

Consider the Ufactory Lite 6 robot arm in the figure below.



It is a six rotational dof serial manipulator that can be controlled from a laptop via an Ethernet cable. Only one laptop at a time can be connected to the robot.

Programming can be done in Python. Scripts available from the course webpage illustrate basic functionalities, namely how to reach a position and how to move a joint with a given velocity. Additional resources can be found at https://ufactory.cc/lite-6-collaborative-robot/ and http://github.com/xArm-Developer/xArm-Python-SDK.

Task 1
Determine the direct kinematics for the Lite 6 robot. Explain, clearly the frame assignment and the method used to obtain the kinematics.

Task 2
Design a trajectory analogue to the swing movement of Lab assignment 1 for the robot to execute, i.e., determine functions of time for the position and orientation of the gripper that correspond to the swing movement (of course adapted to the dimensions of the robot). The ball is fixed, at position arbitrarily chosen, within the reachable space of the robot.

Ensure that the trajectory is feasible by the robot, e.g., the trajectory must not require (i) that the robot moves with a velocity that exceeds the manufacturer specs, (ii) the robot gripper move to a point outside the reachability region, and (iii) the physical limits of the joints are not violated.

Provide clear plots of the trajectory obtained. These should clearly demonstrate that the robot will not make any dangerous physical contact.

Task 3

Using the built-in functions available, develop a Python script to make the robot follow the trajectory obtained in Task 2.

Task 4

For a significant number of points of the trajectory in Task 2 distributed along the full trajectory, obtain the corresponding joint variables. Use the robot built-in Python functions available.

Using these joints variables in the direct kinematics, obtain the corresponding positions and orientations. Discuss the differences, if any, between the trajectories obtained and conclude about the accuracy of the built-in inverse kinematics.